How to use ITK and VTK

As both toolkits consist of C++ libraries, a suitable compiler is required, depending on the platform you use. ITK uses the CMake open source build system, which you can get from: *http://www.cmake.org/*.

To write a simple program using ITK or VTK:

- Make a directory for your program.
- A file named *CmakeLists.txt* has to be made to generate the *Makefile* (if you are using UNIX) or Visual Studio Workspace (if you are using MS-Windows).
- Now create a source file of the extension *.cxx*.
- Configure CMake for *make* for the target directory that contains your code. If you are in the target directory, just use ".", else specify the directory name.
- Build the program you made, and…
- Execute the program.

Contributing to ITK

There can be several levels at which you can contribute to ITK, like new classes, internal modules, bridge modules, etc. Each level has its own method; the details are at *http://www.itk.org/Wiki/ITK_Release_4/New_Code_Contribution_Process*.

Contributing to VTK

The simplest way to contribute is to report bugs via the bug tracker (*http://www.vtk.org/Bug/my_view_page.php*). Before writing and submitting code, one must be well aware of VTK's Quality Dashboard (*http://open.cdash.org/index.php?project=VTK*). To subscribe to the VTK developer mailing list, visit *http://public.kitware.com/mailman/listinfo/VTK-developers*. To subscribe to the VTK user mailing list, visit *http://www.vtk.org/VTK/help/vtkusers.html*. You can find information about VTK coding standards at *http://www.vtk.org/Wiki/VTK_Coding_Standards*. END

References

[1] ITK: *http://www.itk.org/*
[2] VTK: *http://www.vtk.org/*

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Ushahidi

Ushahidi is an open source crowd-sourcing platform that can be used to collate and map information gathered from the public, via email, SMS, Twitter, etc. It has been used around the world to map or monitor incidents such as election violence, earthquakes, and several crisis situations.

The Ushahidi platform is built on the Kohana Web framework. It includes built-in support for the Nexmo wholesale SMS API and Clickatell SMS gateways. Ushahidi provides the option of using OpenStreetMap maps in its user interface, but requires the Google Maps API for geo-coding.

Ushahidi currently has an Android and Windows Mobile app, and plans on releasing an iPhone app later this year. Although apps aren't required for crowd-mapping (SMS makes most sense in places such as Haiti), smartphones make the use of Ushahidi significantly more powerful, allowing automatic loading of GPS coordinates and pictures. Ushahidi enables information to be gathered from the public through a variety of channels. It can be set up relatively quickly, and supports several languages. Once an Ushahidi instance (a running Ushahidi software) is set up, messages can be sent in via email, SMS, Twitter, or directly on the instance website. The messages are verified by an administrator, and once approved, are published—and the location of the incident is displayed on a map. Incidents can be categorised, enabling a visual display of incidents and categories.

The SwiftRiver API and platform

SwiftRiver is a platform comprising APIs for data gathering and extracting useful information. It helps people make sense of a lot of information in a short interval of time. In practice, SwiftRiver enables the filtering and verification of real-time data from channels like Twitter, SMS, email and RSS feeds. It is an API ecosystem. It helps to curate real-time data and analysis on any topic of interest to you or your organisation. It helps users discover nascent relationships and trends in data sets that may appear to be unrelated. It can be used to set up streams that search for mentions of your brand or product online and manage social media campaigns after disasters (e.g., via Twitter, SMS or email) from one dashboard. SwiftRiver adds context to content, using semantic analysis. It auto-categorises and classifies email, Twitter, text messages or news articles based on keywords. END

References

[1] SMS for Life: *http://www.rollbackmalaria.org/psm/smsWhatIsIt.html*
[2] Ushahidi: *http://ushahidi.com/products/ushahidi-platform*
[3] SwiftRiver: *http://ushahidi.com/products/swiftriver-platform*

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